Impacts of Simulated Coastal Ocean Alkalinity Enhancement on the Seasonal Cycle of CO_2 Air-Sea Gas Exchange and Ocean pCO_2 in European Waters under Low- and High-Emission Scenarios

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Key Points:

- Ocean Alkalinity Enhancement (OAE) is a carbon dioxide removal option that aims to increase the ocean potential to sequester atmospheric CO_2
- OAE dampens the ocean pCO_2 seasonal cycle, especially under low warming
- OAE amplifies the CO_2 flux seasonal cycle, especially under high warming
- Regions like the shallow, well-mixed waters may be ideal to implement ocean alkalinity enhancement

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Abstract

To meet the Paris Agreement, emission reduction must be complemented by carbon dioxide removal technologies aiming to sequester atmospheric CO_2 and store it in permanent natural reservoirs. One potentially scalable method is ocean alkalinity enhancement (OAE), which works to lower ocean pCO_2 and accelerate CO_2 sequestration and storage as bicarbonate and carbonate ions. As seasonality is a fundamental component of the ocean net annual CO_2 uptake, this study addresses the knowledge gap on the potential impacts of OAE on the seasonal carbon cycle. By analysing earth system model simulations of OAE performed at the European coastline with different climate change forcing, it was found that: a) with alkalinity addition, the ocean pCO_2 seasonal cycle is dampened due to its decreased sensitivity to CO_2 fluctuations; b) the CO_2 seasonal flux is amplified due to the larger imbalance at the air-sea interface in an alkalinised ocean; c) while the ocean pCO_2 seasonal amplitude reduction is stronger under low warming, the CO_2 flux seasonal amplification is stronger in the high warming pathway.

Plain Language Summary

Although emission reduction remains the main strategy to mitigate human-induced global warming, technologies are being developed to sequester excess carbon dioxide from the air. Among the proposed CO_2 removal options is ocean alkalinity enhancement (OAE), which is a method that involves the addition of alkaline minerals or solutions at the ocean surface to lower the CO_2 concentration in seawater and accelerate its uptake from the atmosphere. As seasonality is important to understand the annual carbon movement, this research helps define the impacts of OAE on the seasonal CO_2 cycle. To do this, a numerical earth system model simulated continuous alkalinity addition at the European coastline from 2015 to 2100, where it was found that adding alkalinity to the ocean's surface reduces its seasonal sensitivity and contributes to take up additional CO_2 from the atmosphere, especially over the winter months.

1 Introduction

1.1 Research context

Owing to its physical, chemical, and biological properties, the ocean constitutes a valuable resource for climate stabilisation, absorbing much of the earth's excess heat and storing vast amounts of carbon in its interior and deep waters (Friedlingstein et al., 2022; Scott-Buechler & Greene, 2019). With such premises, research is expanding on marine Carbon Dioxide Removal (CDR), which consists in the employment of wetlands, bank shelves and the open ocean to enhance natural CO_2 removal processes. One of the marine CDR under investigation is ocean alkalinity enhancement (OAE), which aims to speed up the ocean CO_2 uptake through accelerated chemical weathering, namely the breakdown of minerals by water-induced chemical reactions.

Alkalinity is an ocean property that describes the excess of bases, or proton acceptors, over acids, or proton donors (Wolf-Gladrow et al., 2007). Theoretically, OAE would be performed by releasing alkaline minerals or solutions at the ocean surface. Chemically, this operation would convert aqueous CO_2 (CO_2(aq)) into stable bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}) ions, thus lowering ocean CO_2 partial pressure (pCO_2) and allowing for additional CO_2 uptake from the atmosphere to restore the altered equilibrium. While increasing the ocean buffering capacity, OAE would have the additional benefit of counteracting intensifying ocean acidification through the bonding with free hydrogen ions (H^+) in seawater.

Although OAE has been widely investigated to account for theoretical benefits and challenges, little attention has been given to its effects on the seasonality of the carbon cycle, which is a major component of the annual net ocean uptake. OAE implementation is likely to trigger significant seasonal impacts. For example, changes to the CO_2 flux in one direction might not be counterbalanced by compensating effects in the opposite direction, and these asymmetries could encourage modifications to the annual net cycle (Fassbender et al., 2022). Furthermore, as CO_2 flux seasonality is a central driver of ocean extreme events such as earlier calcium carbonate undersaturation (Kwiatkowski et al., 2022) and intensified ocean acidification (Sasse et al., 2015), other than acting upon inter-annual variability patterns (Rustogi et al., 2023), unknown looping mechanisms may affect the marine ecosystem at different scales. Lastly, while atmospheric pCO_2 responds to system stimuli at much shorter timescales, the ocean can take several seasons to react (Salt et al., 2013) and understanding such time lags is necessary for future OAE application.

This study investigates these dynamics using output data from an Earth System Model (ESM) that simulated OAE along the European coastline under a low-emission (SSP1-2.6)

and a high-emission (SSP3-7.0) pathway. Our analysis aims to (a) define the impacts of OAE on the seasonal cycle of CO_2 flux and ocean pCO_2, and (b) evaluate the influence of the background climate scenario on such outcomes.

2 Methods

2.1 Earth System Model

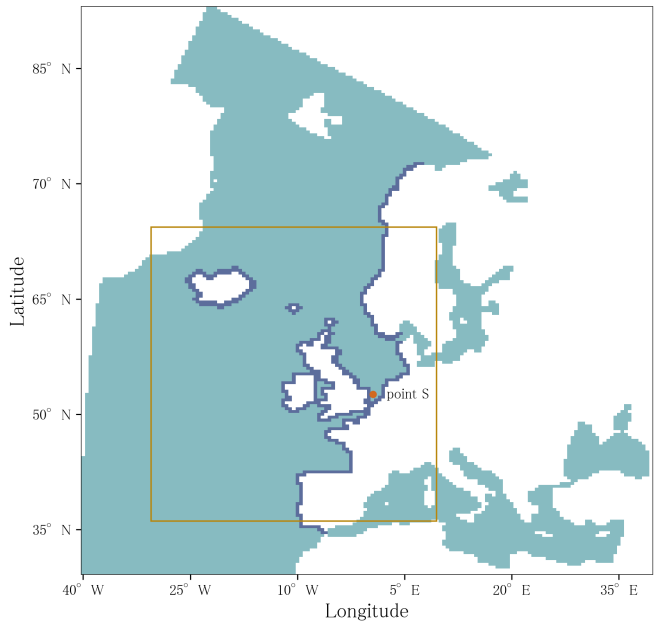
The data for this manuscript was derived from the ESM Flexible Ocean and Climate Infrastructure (FOCI) (Matthes et al., 2020). The model consists of an atmosphere–ocean–sea ice general circulation model with a surface land component. The atmosphere is divided into 95 vertical layers and has a horizontal grid resolution of $\sim$1.8°, while the ocean is composed of 46 vertical layers and has a horizontal resolution of 1/2°.

FOCI incorporates the Model of Oceanic Pelagic Stoichiometry (MOPS) to enable the simulation of marine biogeochemical cycles, including a full carbon cycle (Chien et al., 2022). This allows to resolve biological dynamics such as phytoplankton uptake and remineralisation, calcite formation and dissolution.

2.2 Experimental design

Historical simulations were run from year 1850 to year 2015, when the socio-economic pathway (SSP) begins (Riahi et al., 2017), and both the baseline simulations and forcings follow the CMIP6 protocols (O'Neill et al., 2016). In order to define the role of the future climate change scenario on OAE application, a low- and a high-emission SSP have been selected (SSP1-2.6 and SSP3-7.0, respectively) and simulations were conducted in emission-driven mode (O'Neill et al., 2016).

Figure 1: Geographical OAE application. The dark blue bordering line represents alkalinity addition and the yellow rectangle defines the cropped area used in this paper. Red circle is the location of point S.



OAE is applied from year 2025 to year 2100. Alkalinity addition is subject to a linear increase over the first simulation decade (2025-2035), until the alkalinity equivalent of 1 Gt yr^{-1} of fast-reacting calcium hydroxide ($Ca(OH)_2$) addition is reached.

In FOCI, both dissolved inorganic carbon (DIC) and alkalinity agree well with observations, while still showing a positive bias at the surface and a negative bias below 3000 meters of depth (Chien et al., 2022). Most DIC variations are due to organic matter build-up and CO_2 flux changes, while alkalinity is a prognostic tracer simulated as a combination of biological sources and sinks: nitrate and phosphate supply as sinks, calcium carbonate production and dissolution as sink and source, respectively, and organic matter (OM) formation and remineralisation, as sink and source, respectively (Chien et al., 2022).

$$Ca(OH)_2 + 2CO_2 \leftrightarrow Ca^{2+} + 2HCO_3^- \tag{1}$$

$Ca(OH)_2$ consumes two moles of atmospheric CO_2 and produces one mole of calcium (Ca^{2+}) and two moles of bicarbonate, therefore ideally increasing alkalinity by a factor of two (eq. (1)) (Chien et al., 2022). In our simulations, alkalinity is added to the system continuously and uniformly along a 50-kilometre-wide strip of the European coastline, with the exclusion of the Baltic and the Mediterranean seas (fig. 1).

2.3 Data processing

The global output data delivered by FOCI was sliced to the European region, with coordinates -25° to 10° E and 40° to 70° N (yellow rectangle in fig. 1). Although the cropped area does not cover the entire European region where OAE was performed, and part of the Norwegian and Spanish littoral were left out of the analysis, this approximation does not result in a meaningful modification from the modelled regional definition.

A comparison between the European region and location S set at 52°N and 3°E (red dot in fig. 1) is able, on the one hand, to determine an approximated estimate of how the European system as a whole would behave in the future and, on the other, to draw more precise conclusions on the regime shifts taking place at the epicenter of the domain.

Assuming that the most evident baseline-to-OAE disparities are detected at the end of the simulation, the monthly cycle averaged over the last simulation decade (2090-2100) is provided. While ocean pCO_2 and CO_2 flux are estimated only for the sea surface, alkalinity is measured over all 46 FOCI ocean layers. Thus, alkalinity averages were vertically averaged over the mixed layer depth (MLD), which is the region considered to be in direct contact with the atmosphere and most favourable to CO_2 exchange. The alkalinity that is subducted below the MLD is assumed to not induce CO_2 uptake by the ocean, at least on shorter time scales.

3 Mechanisms driving the CO_2 seasonal flux

3.1 Main drivers of seasonality

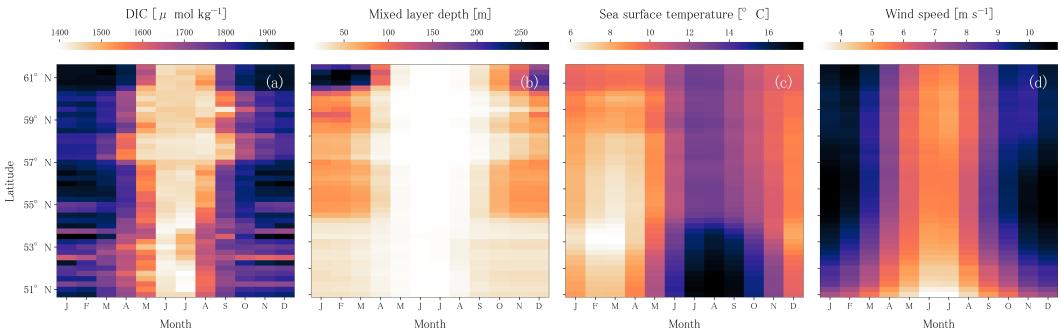
Seasonality is a fundamental property of the annual net carbon cycle (Rodgers, 2023; Fassbender et al., 2022). Although with regional and local variations, ocean CO_2 seasonal flux is mainly controlled by two factors: sea surface temperature (SST) and DIC.

As SST acts on gas dissolution in seawater, in winter months, colder temperatures favour CO_2 solubility while warming water in summer leads to CO_2 outgassing (Williams & Follows, 2011). As for DIC-driven seasonality, in winter, where processes like upwelling and OM respiration prevail and supersaturate the DIC pool, the ocean easily loses CO_2 to the atmosphere, whereas summertime OM production and downwelling favour atmospheric CO_2 drawdown. As the dynamics induced by SST and DIC oppose each other (Lerner et al., 2021; Takahashi et al., 2002), the one prevailing drives the direction of seasonal CO_2 exchange at a given location (Sarmiento & Gruber, 2006).

3.2 Seasonality in the study area

The air-sea CO_2 flux in European waters is characterised by a strong North-to-South seasonal gradient (Salt et al., 2013; Prowe et al., 2009). Figure 2 offers a representation of some of the features that affect CO_2 flux in the North Sea, averaged over the entire simulation period (2015-2100) in SSP1-2.6. Biological drivers dominate in the well stratified northern North Sea (fig. 2a), where a relatively shallow mixed layer allows for nutrient-rich waters to stay at the surface and encourage phytoplankton bloom during the productive season. OM that there accumulates then sinks below the surface layer, where remineralisation can happen far from atmospheric contact. At this location, CO_2 sequestration through carbon fixation outweighs temperature-driven CO_2 outgassing, making the area a strong carbon sink (Salt et al., 2013; Prowe et al., 2009).

Figure 2: From left to right, latitudinal transect of the North Sea (51°N-62°N) for the baseline seasonal cycle of dissolved inorganic carbon (DIC) (a), sea surface temperature (SST) (b), mixed layer depth (MLD) (c), and wind speed (d) averaged over 2015-2100 in SSP1-2.6.



Conversely, in the southern North Sea and the adjacent European continental shelf, where a shallow water column, wind and tidal forces (fig. 2d) generate a more uniform layer all year round (fig. 2c) (Thomas et al., 2009; Hoppema, 1990), the air-sea CO_2 exchange is mainly driven by temperature variability (fig. 2b) (Salt et al., 2013). Carbon production and remineralisation happen within the same surface compartment, where CO_2 can easily be exchanged with the atmosphere (Prowe et al., 2009; Thomas et al., 2004). At this location, community respiration outweighs photosynthetic activity (Artioli et al., 2012), resulting in a region of weak CO_2 outgassing (Prowe et al., 2009).

4 Results

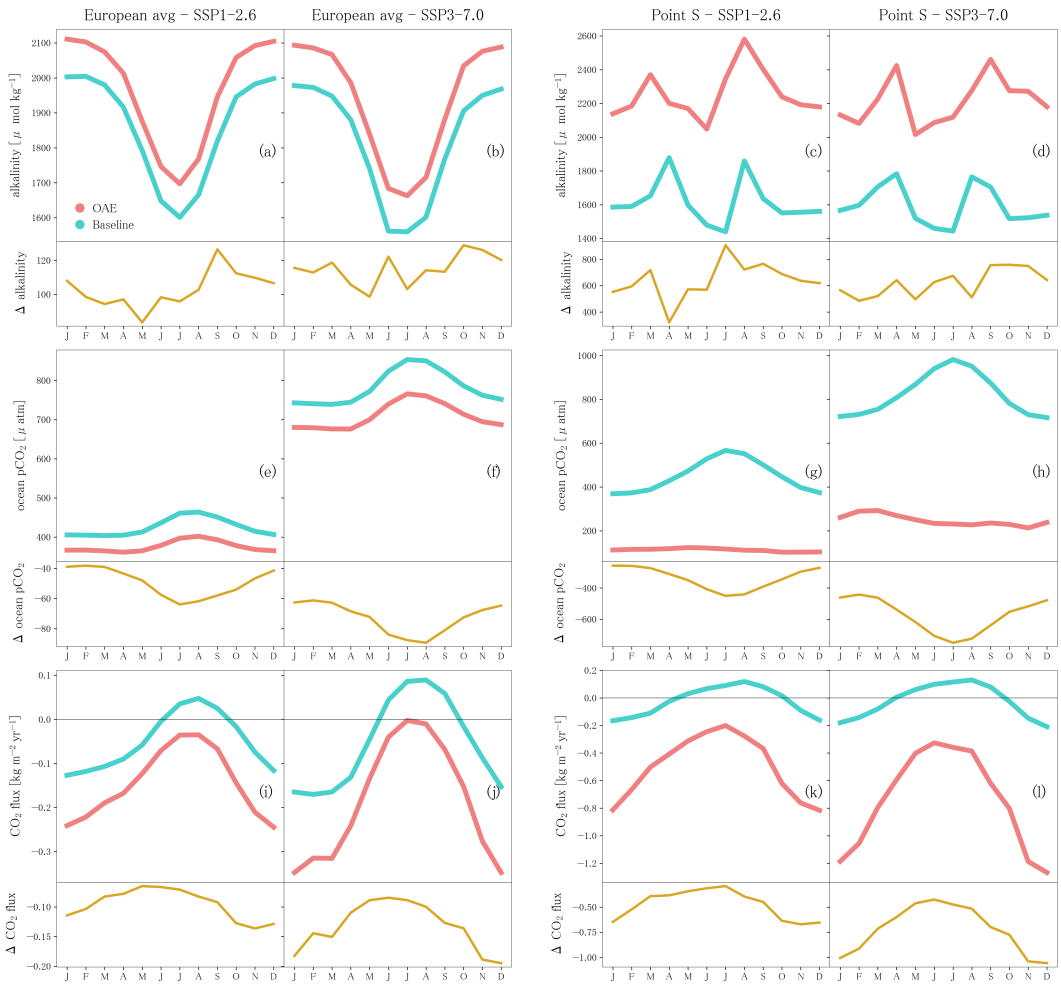
In this section, results are presented for alkalinity, ocean pCO_2, and CO_2 flux. As the geographical maps show, calculations on the CO_2 flux are performed in the atmospheric component of FOCI, which has a coarser horizontal resolution. The lineplots (fig. 3) represent the monthly cycle averaged over the last simulation decade in absolute values, together with the difference between the baseline and the OAE scenario. The maps (fig. 4) display the geographical variation of the seasonal amplitude when OAE is applied. In this manuscript, we define the seasonal amplitude as the difference between the annual maximum and the annual minimum.

4.1 Alkalinity

For the European average in SSP1-2.6 (fig. 3a), baseline alkalinity registers lowest values in summer and highest in winter, with respective extremes corresponding to 1609 and 2008 mmol m^{-3}. With alkalinity addition, absolute values rise at all seasons and the largest change between the two scenarios takes place in September. In SSP3-7.0 (fig. 3b), baseline alkalinity levels are slightly lower than the respective lower-emission pathway but, when OAE is implemented, absolute values rise to similar levels as SSP1-2.6.

At location S for the lower warming scenario, the baseline cycle is similar in both warming trajectories, with two high peaks in April and August, and lowest levels registered in July. Absolute values range between 1439 mmol m^{-3} and 1859 mmol m^{-3} in SSP1-2.6 (fig. 3c), and between 1444 mmol m^{-3} and 1765 mmol m^{-3} in SSP3-7.0 (fig. 3d). Under OAE-induced conditions, absolute values grow substantially and now range from 2048 mmol m^{-3} (June) to 2581 mmol m^{-3} (August) under low warming, and from 2016 mmol m^{-3} (May) to 2460 mmol m^{-3} (September) under high warming.

Figure 3: Annual cycle averaged over 2090-2100 for alkalinity (a, b, c, and d), for ocean pCO_2 (e, f, g, and h), and for the CO_2 flux (i, j, k, and l), with all relative change (baseline - OAE). From left to right, columns indicate the European average in SSP1-2.6 and in SSP3-7.0, and the point S average in SSP1-2.6 and in SSP3-7.0. In CO_2 flux plots, negative values indicate ocean uptake.



In both scenarios, the alkalinity spatial pattern reveals strongest variation in the southern North Sea, where the seasonal amplitude change increases greatly. Almost no difference is observed between SSP1-2.6 (fig. 4a) and SSP3-7.0 (fig. 4d) and no substantial modification is detected in the open ocean.

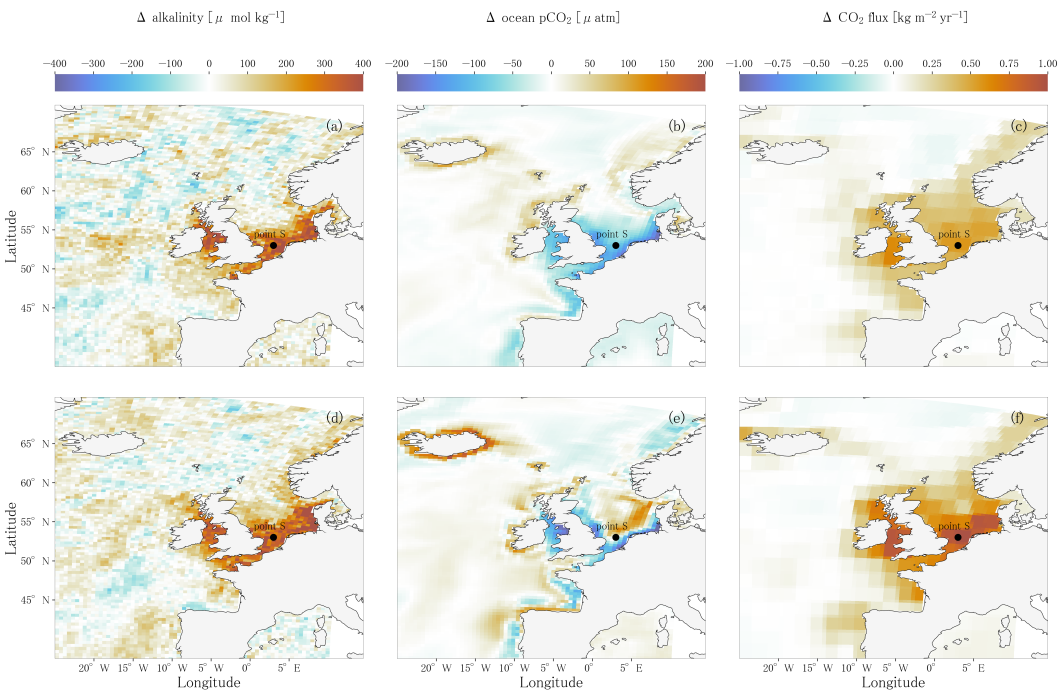
4.2 Ocean pCO_2

For the European average in SSP1-2.6 (fig. 3e), ocean pCO_2 is characterised by least and most elevated values in May and August, respectively, with a mean amplitude stretching from 407 µatm to 441 µatm. In the OAE projection, ocean pCO_2 is reduced due to alkalinity enhancement and absolute values drop. The amplitude also decreases, now spanning from

362 µatm to 385 µatm. Change in ocean pCO_2 shows the largest baseline-to-OAE anomaly in summer, meaning that alkalinity stimulates a pCO_2 decline during CO_2 outgassing. In the baseline of SSP3-7.0 (fig. 3f), pCO_2 maintains an ascending trend until 2100 (not shown). Elevated values are recorded in July, to then decline until minima are reached in April. The mean seasonality has lowest and highest peaks of 744 µatm and 817 µatm, respectively. OAE induces the seasonal cycle to decrease substantially, with an average amplitude of 55 µatm.

Unlike the European average, location S displays ocean pCO_2 minima in winter under both warming scenarios. OAE reduces pCO_2 absolute values, with a mean drop of 337 µatm in SSP1-2.6 (fig. 3g) and of 574 µatm in SSP3-7.0 (fig. 3h). OAE contributes to ocean pCO_2 decrease especially over the warm months, with a significant seasonal amplitude reduction: in the low warming scenario, it is compressed by more than a third compared to the baseline, shifting from 197 µatm to 21 µatm, while in SSP3-7.0 it is almost halved, moving from 266 µatm to 79 µatm.

Figure 4: Seasonal amplitude change (baseline - OAE) averaged over 2090-2100 for alkalinity (a and d), for ocean pCO_2 (b and e) and for the CO_2 flux (c and f). The top row is scenario SSP1-2.6 and the bottom row is scenario SSP3-7.0. The black circle is point S.



In SSP1-2.6 (fig. 4b), a clear spatial pattern is identified at the proximity of the injection location, with two diverging geographies: at higher latitudes, in Iceland and in the

Norwegian and British North-West bank, the pCO_2 seasonal amplitude is slightly more pronounced than the baseline. Conversely, at lower latitudes including the mainland shoreline and the central-to-southern UK coasts, the pCO_2 seasonal cycle is strongly mitigated.

In SSP3-7.0 (fig. 4e), the pCO_2 amplitude behaves similarly and maintains the spatial pattern outlined above, with the exception of the central North Sea, where strong amplification is displayed. In the open sea, pCO_2 seasonality is slightly amplified in the North Atlantic and dampened in the sub-polar regions. Iceland represents a deviation from the expected outcome, as it displays strong seasonal amplification with OAE application.

4.3 CO_2 flux

Figure 3i shows that, for the European average in SSP1-2.6, the baseline ocean tends to outgass CO_2 in summer (July and August), while remaining a carbon sink over the remaining months, with most pronounced uptake in winter, especially December. When OAE is applied, the system turns into a carbon sink all year round, with summertime (wintertime) still reflecting the weakest (strongest) CO_2 uptake. The seasonal amplitude varies noticeably, with the OAE scenario showing a much greater breadth than the baseline. OAE-induced CO_2 flux amplitude has an average span of -0.0349 kg m^{-2} yr^{-1} to -0.2446 kg m^{-2} yr^{-1}, compared to the baseline that ranged between 0.0474 kg m^{-2} yr^{-1} and -0.1268 kg m^{-2} yr^{-1}. In SSP3-7.0 (fig. 3j), the baseline outgasses CO_2 from July to October. Adding alkalinity increases the CO_2 seasonal amplitude more than the low emission trajectory, especially over the cold months.

In the baseline of both scenarios at location S (fig. 3k, l), the annual cycle is split into two equally long phases. Outgassing happens from May to October (SSP1-2.6) and October (SSP3-7.0) and uptake from November to March (SSP1-2.6) and April (SSP3-7.0). In OAE-driven SSP1-2.6, CO_2 flux fluctuates between -0.8156 kg m^{-2} yr^{-1} and -0.2004 kg m^{-2} yr^{-1} on average, therefore doubling the baseline amplitude, while seasonality of CO_2 flux in SSP3-7.0 is three times larger than the baseline, with a current amplitude of 0.9416 kg m^{-2} yr^{-1}. In all Δ CO_2 graphs, the baseline-to-OAE discrepancies grow in winter and decrease in summer, meaning that the impacts of alkalinisation on the ocean surface are largest when ocean uptake is most pronounced.

With OAE, SSP1-2.6 (fig. 4c) reveals a pronounced increase in the CO_2 seasonal flux all around the European continental coastline, and the signal remains strong until the connection with the open ocean. Little-to-no changes are observed in Iceland and Northern Spain. SSP3-7.0 (fig. 4f) mirrors the same spatial pattern as SSP1-2.6, although the North-West Icelandic coast reveals slightly stronger amplification.

5 Discussion

5.1 European average

All maps show that system perturbations for ocean pCO_2 and the CO_2 flux happen within the model domain, especially at the site in proximity of alkalinity injection, and the signal does not propagate far by the end of the century. This result is however not enough to assume that OAE is most successful near the addition site, as this paper does not address alkalinised water that has been subducted, and uncertainty remains on whether that water could resurface somewhere else to complete the process.

While open-ocean regions like Iceland and Spain show few-to-absent perturbations with alkalinity addition, the most affected area seems to be the North Sea, especially the South which delivers pronounced system alterations. This may be due to the fact the shallower, non-stratified ocean system allows for alkalinity to accumulate at the surface and to not be lost due to mixing-induced subduction before equilibration is completed (Wang et al., 2023; Jones et al., 2014).

In both SSP1-2.6 and SSP3-7.0, ocean pCO_2 minima are registered in spring, which may be the result of phytoplankton bloom and consequent high photosynthetic activity, where CO_2 is consumed (Gypens et al., 2011). In SSP3-7.0, substantial atmospheric CO_2 emissions force pCO_2 to rise progressively until the end of the simulation, even with alkalinity enhancement. With OAE application, ocean pCO_2 seasonal amplitude is reduced because an alkalinised ocean minimises pCO_2 sensitivity to DIC changes.

Remarkably, regions like Iceland and Northern United Kingdom constitute an exception, where alkalinity addition causes an amplification signal of the pCO_2 seasonal cycle. This may be due to the diverging reaction of alkalinity addition between a semi-enclosed space and the open ocean, where different processes occur.

In agreement with previous estimates (Schwinger, 2022), OAE works to amplify the CO_2 seasonal flux in both SSPs. This happens because OAE increases the air-to-sea disequilibrium, which thus requires higher CO_2 input into the ocean to restore it. In both scenarios, highest levels of alkalinity are recorded between the end of summer and the onset of autumn but largest uptake takes place in winter, therefore introducing a short response lag due to re-equilibration timescales. In the OAE scenario, alkalinity addition prompts a greater CO_2 flux downward in wintertime, meaning that OAE induces an asymmetrical regime shift, with much lower lows than lower highs.

5.2 Point S

Air-sea equilibration can take from a few months to even years to complete, depending on the physical properties of the region (Jones et al., 2014). Considering the topographic characteristics of the southern North Sea, where a shallow water column combines with strong wind-driven forces and tidal currents, a region of uniform mixing could encourage fast re-equilibration with the atmosphere and allow for efficient carbon sequestration under enhanced alkalinity.

With OAE deployment, pCO_2 levels plummet and OAE-induced pCO_2 change is found to be highest in the season where the associated seasonal driver operates to reduce the ocean CO_2 sink potential. In this case, pCO_2 deviation from the baseline increases in summer, when rising temperatures inhibit ocean uptake.

At location S, CO_2 flux modulations are dominated by temperature changes (Rodgers, 2023; Salt et al., 2013; Prowe et al., 2009). Strongest CO_2 outgassing (and lowest uptake in the OAE-induced scenario) takes place in summer, when warm temperatures weaken solubility, whereas CO_2 uptake happens in winter. Stronger simulated global warming results in higher CO_2 ocean uptake as well as larger seasonal amplification caused by the enhanced air pCO_2 that forces the gas movement into the ocean.

5.3 Influence of the background climate scenario

When analysing the influence of the climate pathway, ocean pCO_2 and the CO_2 flux have opposite reactions. As for the former, though absolute values show a less significant decline in SSP1-2.6, the compression signal is much stronger in this scenario. Conversely, the seasonal amplification of the CO_2 flux is exacerbated under higher warming.

Two key insights emerge from these results. First, the sensitivity of oceanic pCO_2 increases with more intense global warming, which suggests that OAE alone is insufficient to effectively mitigate changes to ocean pCO_2 seasonality. Second, rather than alleviating the issue, OAE actually intensifies a pre-existing, temperature-driven phenomenon—namely, the amplification of seasonal fluctuations in CO_2 caused by rising temperatures.

6 Conclusions

This study addressed the OAE-induced changes on the seasonal carbon cycle in European waters, therefore contributing to the understanding of CO_2 intra-annual variability when performing coastal alkalinity enhancement. It was found that: (a) OAE dampens the ocean pCO_2 seasonal cycle and it amplifies the CO_2 flux cycle; (b) the former outcome is stronger in SSP1-2.6 while the latter turns stronger in SSP3-7.0; (c) locations like the southern North Sea could be ideal to implement OAE, as a shallow water column and intense mixing can fasten the CO_2 gas transfer at the air-sea interface, at least as far as the requilibration time scale is shorter than the surface water mean residence time.

Some limitations and recommendations for future research can be drawn from this analysis. Firstly, while freshwater input from rivers is modelled in FOCI, riverine alkalinity is not accounted for, limiting the reliability of resulting alkalinity absolute values. As river runoff features are deeply seasonal, implementing such component into future simulations would add complexity and precision to the system, so to enhance the understanding of external sources to the European seas.

Secondly, our analysis does not separate ocean pCO_2 into its thermal and biological element, therefore missing to discern the influence that each of these have in altering the CO_2 seasonal cycle. Given the significance of temperature variability in the southern North Sea, taking each of the two terms individually could provide further insight into their seasonal role under OAE deployment.

Lastly, our OAE simulations pictures a highly idealised scenario, which is unlikely to become a real-world case. Other, more realistic OAE techniques should be explored, such as alkalinity addition performed in pulses or at selected point locations (river mouths, sewage treatment plants, open ocean regions). To conclude, as the field of carbon cycle seasonal variability within OAE is still largely unexplored, further investigation is recommended to build the premises for conscious large-scale OAEapplication.

7 Author contribution

Conceptualisation: C. Ciscato, D. P. Keller; Data curation: C. Ciscato, D. P. Keller; Formal analysis: C. Ciscato; Investigation: C. Ciscato; Methodology: D. P. Keller; Supervision: D. P. Keller; Validation: D. P. Keller; Visualisation: C. Ciscato; Writing – original draft: C. Ciscato; Writing – review & editing: C. Ciscato, D. P. Keller, Mehendale, N.

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9 Open Research

The code underlying this study is openly available at the following GitHub repository: 2024_ciscato_oae_seasonality

References

Artioli, Y., Blackford, J. C., Butenschön, M., Holt, J. T., Wakelin, S. L., Thomas, H., ... Allen, J. I. (2012). The carbonate system in the north sea: Sensitivity and model validation. *Journal of Marine Systems*, *102*, 1–13.

Chien, C.-T., Durgadoo, J. V., Ehlert, D., Frenger, I., Keller, D. P., Koeve, W., ... others (2022). Foci-mops v1–integration of marine biogeochemistry within the flexible ocean and climate infrastructure version 1 (foci 1) earth system model. *Geoscientific Model Development Discussions*, 1–58.

Fassbender, A. J., Schlunegger, S., Rodgers, K. B., & Dunne, J. P. (2022). Quantifying the role of seasonality in the marine carbon cycle feedback: an esm2m case study. *Global Biogeochemical Cycles*, *36*(6), e2021GB007018.

Friedlingstein, P., O'Sullivan, M., Jones, M. W., Andrew, R. M., Gregor, L., Hauck, J., ... others (2022). Global carbon budget 2022. *Earth System Science Data*, *14*(11), 4811–4900.

Gypens, N., Lacroix, G., Lancelot, C., & Borges, A. V. (2011). Seasonal and inter-annual variability of air–sea co2 fluxes and seawater carbonate chemistry in the southern north sea. *Progress in oceanography*, *88*(1-4), 59–77.

Hoppema, J. (1990). The distribution and seasonal variation of alkalinity in the southern

bight of the north sea and in the western wadden sea. *Netherlands journal of sea research*, *26*(1), 11–23.

Jones, D. C., Ito, T., Takano, Y., & Hsu, W.-C. (2014). Spatial and seasonal variability of the air-sea equilibration timescale of carbon dioxide. *Global Biogeochemical Cycles*, *28*(11), 1163–1178.

Kwiatkowski, L., Torres, O., Aumont, O., & Orr, J. C. (2022). Modified future diurnal variability of the global surface ocean co2 system. *Global Change Biology*.

Lerner, P., Romanou, A., Kelley, M., Romanski, J., Ruedy, R., & Russell, G. (2021). Drivers of air-sea co2 flux seasonality and its long-term changes in the nasa-giss model cmip6 submission. *Journal of Advances in Modeling Earth Systems*, *13*(2), e2019MS002028.

Matthes, K., Biastoch, A., Wahl, S., Harlaß, J., Martin, T., Brücher, T., ... others (2020). The flexible ocean and climate infrastructure version 1 (foci1): mean state and variability. *Geoscientific Model Development*, *13*(6), 2533–2568.

O'Neill, B. C., Tebaldi, C., Van Vuuren, D. P., Eyring, V., Friedlingstein, P., Hurtt, G., ... others (2016). The scenario model intercomparison project (scenariomip) for cmip6. *Geoscientific Model Development*, *9*(9), 3461–3482.

Prowe, A. F., Thomas, H., Pätsch, J., Kühn, W., Bozec, Y., Schiettecatte, L.-S., ... de Baar, H. J. (2009). Mechanisms controlling the air–sea co2 flux in the north sea. *Continental Shelf Research*, *29*(15), 1801–1808.

Riahi, K., Van Vuuren, D. P., Kriegler, E., Edmonds, J., O'neill, B. C., Fujimori, S., ... others (2017). The shared socioeconomic pathways and their energy, land use, and greenhouse gas emissions implications: An overview. *Global environmental change*, *42*, 153–168.

Rodgers, K. (2023). *Seasonal variabiltiy of the surface ocean carbon cycle: a global synthesis* (Tech. Rep.). Copernicus Meetings.

Rustogi, P., Landschützer, P., Brune, S., & Baehr, J. (2023). The impact of seasonality on the annual air-sea carbon flux and its interannual variability. *npj Climate and Atmospheric Science*, *6*(1), 66.

Salt, L. A., Thomas, H., Prowe, A. F., Borges, A. V., Bozec, Y., & de Baar, H. J. (2013). Variability of north sea ph and co2 in response to north atlantic oscillation forcing. *Journal of Geophysical Research: Biogeosciences*, *118*(4), 1584–1592.

Sarmiento, J., & Gruber, N. (2006). *Ocean biogeochemical dynamics*. Princeton University Press.

Sasse, T., McNeil, B., Matear, R., & Lenton, A. (2015). Quantifying the influence of co 2 seasonality on future aragonite undersaturation onset. *Biogeosciences*, *12*(20), 6017–6031.

Schwinger, J. (2022). Report on modifications of ocean carbon cycle feedbacks under ocean alkalinization.

Scott-Buechler, C. M., & Greene, C. H. (2019). Role of the ocean in climate stabilization. *Bioenergy with Carbon Capture and Storage*, 109–130.

Takahashi, T., Sutherland, S. C., Sweeney, C., Poisson, A., Metzl, N., Tilbrook, B., ... others (2002). Global sea–air co2 flux based on climatological surface ocean pco2, and seasonal biological and temperature effects. *Deep Sea Research Part II: Topical Studies in Oceanography*, *49*(9-10), 1601–1622.

Thomas, H., Bozec, Y., Elkalay, K., & De Baar, H. J. (2004). Enhanced open ocean storage of co2 from shelf sea pumping. *Science*, *304*(5673), 1005–1008.

Thomas, H., Schiettecatte, L.-S., Suykens, K., Koné, Y., Shadwick, E., Prowe, A. F., ... Borges, A. (2009). Enhanced ocean carbon storage from anaerobic alkalinity generation in coastal sediments. *Biogeosciences*, *6*(2), 267–274.

Wang, H., Pilcher, D. J., Kearney, K. A., Cross, J. N., Shugart, O. M., Eisaman, M. D., & Carter, B. R. (2023). Simulated impact of ocean alkalinity enhancement on atmospheric co2 removal in the bering sea. *Earth's Future*, *11*(1), e2022EF002816.

Williams, R. G., & Follows, M. J. (2011). *Ocean dynamics and the carbon cycle: Principles and mechanisms*. Cambridge University Press.

Wolf-Gladrow, D. A., Zeebe, R. E., Klaas, C., Körtzinger, A., & Dickson, A. G. (2007). Total alkalinity: The explicit conservative expression and its application to biogeochemical processes. *Marine Chemistry*, *106*(1-2), 287–300.